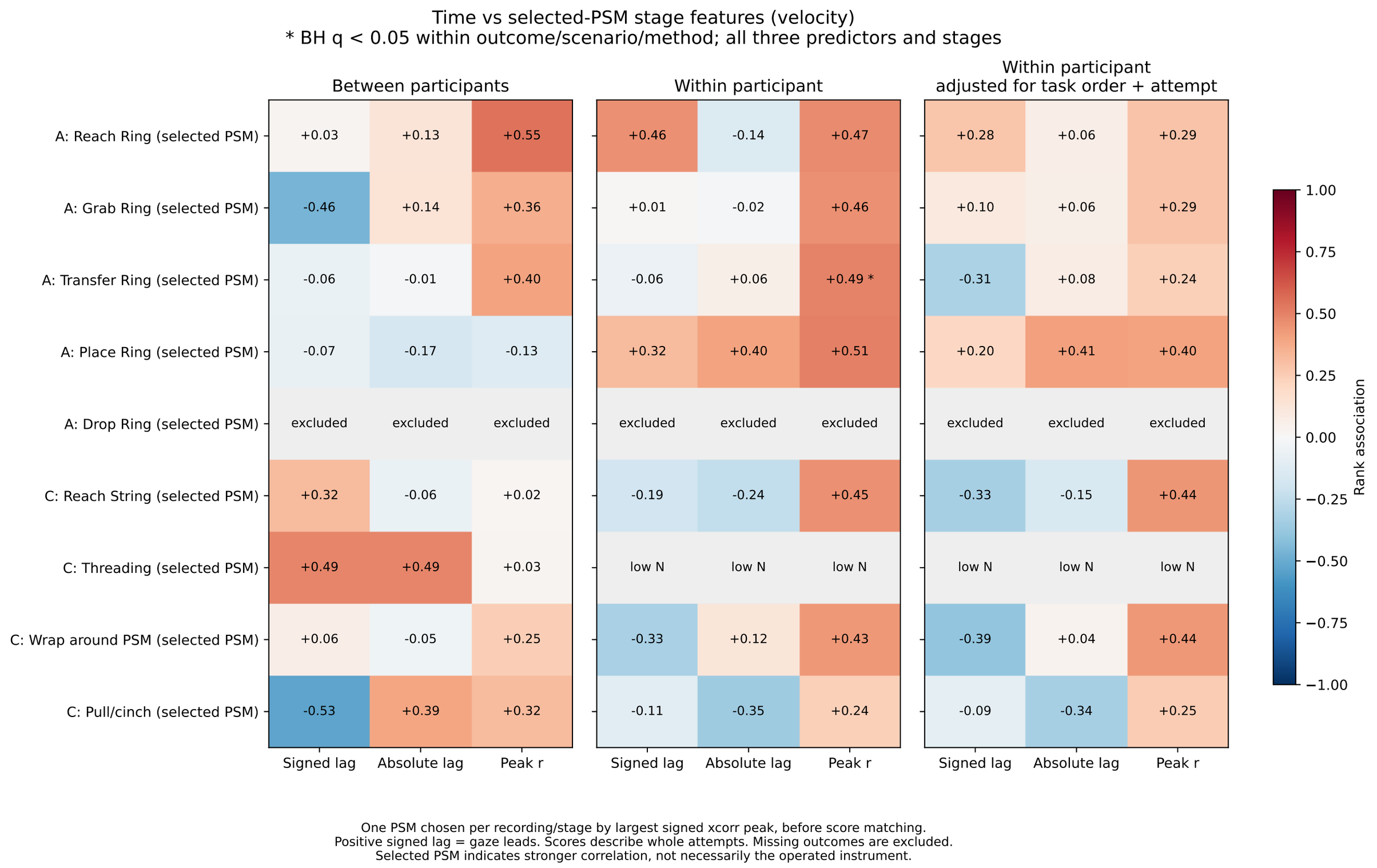
Time vs selected-PSM stage features (velocity)
* BH q < 0.05 within outcome/scenario/method; all three predictors and stages
Between participants
Within participant
Within participant
adjusted for task order + attempt
A: Reach Ring (selected PSM)
A: Grab Ring (selected PSM)
A: Transfer Ring (selected PSM)
A: Place Ring (selected PSM)
A: Drop Ring (selected PSM)
C: Reach String (selected PSM)
C: Threading (selected PSM)
C: Wrap around PSM (selected PSM)
C: Pull/cinch (selected PSM)
+0.03 +0.13 +0.55
-0.46 +0.14 +0.36
-0.06 -0.01 +0.40
-0.07 -0.17 -0.13
excluded excluded excluded
+0.32 -0.06 +0.02
+0.49 +0.49 +0.03
+0.06 -0.05 +0.25
-0.53 +0.39 +0.32
+0.46 -0.14 +0.47
+0.01 -0.02 +0.46
-0.06 +0.06 +0.49 *
+0.32 +0.40 +0.51
excluded excluded excluded
-0.19 -0.24 +0.45
low N low N low N
-0.33 +0.12 +0.43
-0.11 -0.35 +0.24
+0.28 +0.06 +0.29
+0.10 +0.06 +0.29
-0.31 +0.08 +0.24
+0.20 +0.41 +0.40
excluded excluded excluded
-0.33 -0.15 +0.44
low N low N low N
-0.39 +0.04 +0.44
-0.09 -0.34 +0.25
Signed lag
Absolute lag
Peak r
Rank association
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.